

CODE ALPHA INTERNSHIP

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C++ Programming Tasks:

TASK NO. 03: *Banking System*

Design classes for Customer, Account, and Transaction. Implement features like creating accounts, deposits, withdrawals, fund transfers, storing transaction history, and displaying account balance & transaction details.

Code:

```
Banking System.cpp ×
1  #include <iostream>
2  #include <vector>
3  #include <string>
4  #include <iomanip>
5  using namespace std;
6
7  class Transaction {
8  public:
9      string type;
10     double amount;
11     string date;
12     Transaction(string t, double a, string d) : type(t), amount(a), date(d) {}
13 };
14
15 class Account {
16 private:
17     string accountNumber;
18     double balance;
19     vector<Transaction> transactions;
20
21 public:
22     Account(string accNum) : accountNumber(accNum), balance(0.0) {}
23
24     void deposit(double amount) {
25         balance += amount;
26         transactions.push_back(Transaction("Deposit", amount, "2026-05-02"));
27         cout << "Deposited: Rs." << amount << endl;
28     }
29
30     bool withdraw(double amount) {
31         if(amount > balance) {
32             cout << "Insufficient balance!\n";
33             return false;
34         }
```

BankingSystem.cpp ×

```
34     }
35     balance -= amount;
36     transactions.push_back(Transaction("Withdrawal", amount, "2026-05-02"));
37     cout << "Withdrawn: Rs." << amount << endl;
38     return true;
39 }
40
41 void showBalance() const {
42     cout << "Account: " << accountNumber << " | Balance: Rs." << fixed << setprecision(2) << balance << endl;
43 }
44
45 void showTransactions() const {
46     cout << "\nTransaction History:\n";
47     for(const auto& t : transactions) {
48         cout << t.date << " | " << t.type << " | Rs." << t.amount << endl;
49     }
50 }
51 };
52
53 int main() {
54     string name, accNum;
55     cout << "=== Banking System ===\n";
56     cout << "Enter your name: ";
57     getline(cin, name);
58     cout << "Enter account number: ";
59     cin >> accNum;
60
61     Account acc(accNum);
62     int choice;
63     double amount;
64
65     while(true) {
66         cout << "\n1. Deposit\n2. Withdraw\n3. Check Balance\n4. Transaction History\n5. Exit\n";
67         cout << "Enter choice: ";
```

```

68         cin >> choice;
69
70     switch(choice) {
71     case 1:
72         cout << "Enter amount: ";
73         cin >> amount;
74         acc.deposit(amount);
75         break;
76     case 2:
77         cout << "Enter amount: ";
78         cin >> amount;
79         acc.withdraw(amount);
80         break;
81     case 3:
82         acc.showBalance();
83         break;
84     case 4:
85         acc.showTransactions();
86         break;
87     case 5:
88         cout << "Thank you for using our bank!\n";
89         return 0;
90     default:
91         cout << "Invalid choice!\n";
92     }
93 }
94

```

OUTPUT:

```

C:\Users\User\Documents\Ba... X + ~
=== Banking System ===
Enter your name: Laiba
Enter account number: 4357684787652

1. Deposit
2. Withdraw
3. Check Balance
4. Transaction History
5. Exit
Enter choice: 1
Enter amount: 20000
Deposited: Rs.20000

1. Deposit
2. Withdraw
3. Check Balance
4. Transaction History
5. Exit
Enter choice: 2
Enter amount: 1000
Withdrawn: Rs.1000

1. Deposit
2. Withdraw
3. Check Balance
4. Transaction History
5. Exit
Enter choice: 3
Account: 4357684787652 | Balance: Rs.19000.00

```

1. Deposit
2. Withdraw
3. Check Balance
4. Transaction History
5. Exit

Enter choice: 4

Transaction History:

2026-05-02 | Deposit | Rs.20000.00

2026-05-02 | Withdrawal | Rs.1000.00

1. Deposit
2. Withdraw
3. Check Balance
4. Transaction History
5. Exit

Enter choice: 5

Thank you for using our bank!

Process exited after 60.84 seconds with return value 0

Press any key to continue . . . |